Bend Complexity and Hamiltonian Cycles in Grid Graphs

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Abstract. Let G be an $m \times n$ rectangular grid graph. We study the problem of transforming Hamiltonian cycles on G under two basic operations we call *flip* and *transpose*. We introduce a new complexity measure, the *bend complexity*, for Hamiltonian cycles. Given any two Hamiltonian cycles C_1 and C_2 of bend complexity 1, we show that C_1 can be transformed to C_2 using only a linear number of flips and transposes.

1 Introduction

A *grid graph* is a finite, embedded, vertex-induced subgraph of the infinite two dimensional integer grid. Its vertex set is finite and its vertices have integer coordinates. Two vertices are adjacent if and only if they are distance one apart. A *solid grid graph* is a grid graph without holes, i.e., each bounded face of the graph is a unit square. An $m \times n$ *rectangular grid graph* is a solid grid graph of area $(m-1) \times (n-1)$ such that the boundary of the outer face is rectangular. This graph has m rows and n columns.

The Hamiltonian path and cycle problems have been studied in detail in the context of grid graphs as initiated by Itai *et al.* in 1982 [8]. They showed that it is NP-complete to decide whether a grid graph has a Hamiltonian path or a Hamiltonian cycle. They also gave necessary and sufficient conditions for a rectangular grid graph to have a Hamiltonian cycle. They left the problem of deciding whether a Hamiltonian cycle exists in a solid grid graph open. Later Umans and Lenhart [9] gave a polynomial time algorithm to find a Hamiltonian cycle (if it exists) in a solid grid graph. Afrati [1] gave a linear time algorithm for finding Hamiltonian cycles in *restricted grids*. Cho and Zelikovsky [4] studied *spanning closed trails* containing all the vertices of a grid graph. They showed that the problem of finding such a trail is NP-complete for grid graphs, but solvable for a subclass of grid graphs that they called *polyminos*. Researchers have explored other grids as well. In 2009, Arkin *et al.* [2] studied the existence of Hamiltonian cycles in grid graphs and proved several complexity results for a variety of types of grids.

In this paper, we define two simple "local" operations, *flip* and *transpose*, and study transformation of one Hamiltonian cycle of a rectangular grid graph into another Hamiltonian cycle under those two operations. The idea of applying local operations to transform one cycle into another was inspired by the study of

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transforming triangulations by "flipping" one edge at a time. In 1936, Wagner showed that any triangulation can be transformed into a canonical form, and hence can be transformed into any other triangulation [10]. Researchers since have studied different aspects of this problem (see [3] for a survey).

Our work is motivated also by applications of Hamiltonian paths and cycles in grid graphs. Grid graphs are used in path planning problems [7]. Suppose a vacuum cleaning robot is cleaning an indoor environment mapped as a grid graph, and to prevent damage to the floor, the robot must visit each vertex exactly once. The problem gives rise to a Hamiltonian path or cycle problem [6]. Another application is exploring an area with a given map. By overlaying the map with a grid graph, exploration is reduced to a Hamiltonian cycle problem. Enumeration of Hamiltonian paths and cycles in grid graphs has found application in polymer science [5].

From now on, unless specified otherwise, the term $m \times n$ *grid graph*, or simply *grid graph*, refers to a *rectangular grid graph*.

The rest of this paper is organized as follows. In Sect. 2, we introduce two operations, *flip* and *transpose*, and a complexity measure for Hamiltonian cycles in rectangular grid graphs that we call *bend complexity*. Section 3 gives an algorithm that, given any two Hamiltonian cycles C_1 and C_2 with bend complexity 1, transforms C_1 into C_2 using only a linear number of flips and transposes. Section 4 concludes the paper with some open problems.

2 Preliminaries

In this section, we introduce two local operations on Hamiltonian cycles on grid graphs that we call *flip* and *transpose*, and a complexity measure for such cycles that we call the *bend complexity*.

Let G be an $m \times n$ grid graph. We assume that G is embedded on the integer grid such that the top left corner vertex is at $(0, 0)$. We also assume that the positive y direction is downward. A vertex of G with integer coordinates (x, y) is denoted as $v_{x,y}$. Vertices with the same x-coordinates or y-coordinates form a *column* or a *row*, respectively. All the vertices of *column* a have x-coordinate a, and all the vertices of *row* b have y-coordinate b. We call a subgraph G' of G a *subgrid* of G if G' is a grid graph, i.e., G' is an $m' \times n'$ rectangular grid graph.

We now define, by way of concrete examples, two "local" operations on any Hamiltonian cycle on G, where by *local*, we mean that the changes to the cycle are made within a small subgrid G' of G.

Flip. Let G' be a 3×2 or a 2×3 subgrid of a grid graph G, where the vertices a, b, d, f, e, c appear on the boundary of the outer face of G' in clockwise order, and the corner vertices of G' are a, b, f, e. Let C be a Hamiltonian cycle of G such that only the edges $(a, c), (c, d), (d, b)$ and (e, f) of G' are included in C. A *flip* operation on C in G' is performed in two steps. First, we remove the edges $(a, c), (c, d)$ and (d, b) and connect a to b, which gives a Hamiltonian cycle on $G - \{c, d\}$. We then remove the edge (e, f) and add the edges $(e, c), (c, d)$

and (d, f), which gives a Hamiltonian cycle C' on G. See Fig. 1(a). Note that applying flip to C' in G' gives back the original Hamiltonian cycle C.

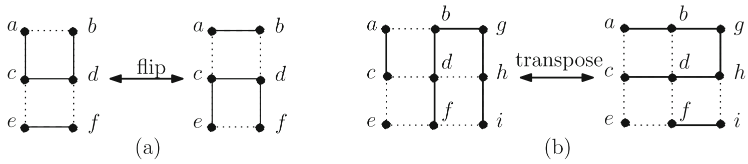

Fig. 1. Two local operations on a Hamiltonian cycle on a grid graph.

Transpose. Let G'' be a 3×3 subgrid of a grid graph G, where the vertices a, b, g, h, i, f, e, c appear on the boundary of the outer face of G'' in clockwise order, and the corner vertices of G'' are a, g, i, e. The vertex d of G'' is the only vertex that does not appear on the boundary of the outer face of G''. Let C be a Hamiltonian cycle on G such that the edges $(a, c), (f, d), (d, b), (b, g), (g, h), (h, i)$ are included in C. Either or both or none of the edges (e, c) and (e, f) might be on C. A *transpose* operation on C in G'' is performed in two steps. First, we remove the edges $(f, d), (d, b), (b, g), (g, h), (h, i)$ and add the edge (f, i). This gives a Hamiltonian cycle in $G - \{d, b, g, h\}$. We then remove the edge (a, c) and add the edges $(a, b), (b, g), (g, h), (h, d), (d, c)$, which gives a Hamiltonian cycle C' in G. See Fig. 1(b). The edges incident to e are not affected by the transpose operation. We call e the *pin vertex* of G''. Note that C' can be transformed back to C by applying transpose on C' in G''.

We introduce a complexity measure for Hamiltonian cycles in G.

Bend Complexity. Let C be a Hamiltonian cycle of G. Let v be a vertex of G and let e_1 and e_2 be the two edges incident to v that are also included in C. We say that v is a *bend* on C if one of e_1, e_2 is horizontal and the other is vertical. We define the *level of a bend* on C as follows. If the bend is on the boundary of the outer face of G, then its level is 0. Otherwise, if the bend is connected to a bend of level k by a straight line of edges of C then the level of this bend is $k + 1$. If the bend is connected to two bends, its level is one more than the minimum of the levels of the neighboring bends.

The *bend complexity* $B(C)$ of Hamiltonian cycle C is the highest level of any bend on C. We call a Hamiltonian cycle with bend complexity k a *k-complex Hamiltonian cycle*. The 1-complex Hamiltonian cycles have simple structures that look like "combs"; cycles with higher bend complexity are more complex.

3 1-Complex Hamiltonian Cycles

In this section, we study the structure of Hamiltonian cycles with bend complexity one, i.e., the 1-complex Hamiltonian cycles. We show that any 1-complex Hamiltonian cycle can be transformed to any other 1-complex Hamiltonian cycle using a linear number of flips and transposes.

We first define some terminology. Let G be an $m \times n$ rectangular grid graph. Throughout this section, we assume that $m, n \geq 3$ as otherwise G has at most one Hamiltonian cycle. The vertices of G with x-coordinate 0 form the *west boundary*, the vertices with x-coordinate $n-1$ form the *east boundary*, the vertices with y-coordinate 0 form the *north boundary*, and the vertices with y-coordinate $m-1$ form the *south boundary.*

Let C be a 1-complex Hamiltonian cycle of G. A *cookie* c is a path in C that contains more than two vertices and that starts and ends at two adjacent vertices on the same boundary (north, south, west or east) of G but does not contain any boundary edges; thus the path travels along a row or column, bends before reaching the opposite side, then immediately bends again to return to the same boundary, ending at a vertex adjacent to the start vertex. Thus, C consists of a set of cookies, and a set of sequences of vertices on C that form straight line segments on the boundary connecting two cookies or a cookie to a corner vertex. We call the length of the parallel sides of c the *height* of c. Observe that the height of a cookie is always greater than zero. Based on which one of the four boundaries contains the endpoints of a cookie, we have four *types* of cookies: north, south, west and east.

A cookie c lies in a pair of adjacent rows (e.g., rows y and $y+1$, for $0 < y < m-2$, when c is a west or east cookie) or adjacent columns (e.g., columns x and $x+1$, for $0 < x < n-2$, when c is a north or a south cookie). We call a pair of adjacent rows of G a *horizontal track* and denote it by t_x, and we call a pair of adjacent columns a *vertical track* and denote it by t_y, where x and y are the smaller indices of the pairs.

If C has only one type of cookie, then we call C a *canonical Hamiltonian cycle*. Note that if G has a Hamiltonian cycle, at least one of m and n must be even [8]. If only one of them is even, then G has two canonical Hamiltonian cycles; otherwise G has four canonical Hamiltonian cycles. Let C_1 and C_2 be any two 1-complex Hamiltonian cycles of G. Let $\mathbb{C}_1$ and $\mathbb{C}_2$ be two canonical Hamiltonian cycles of G. We show that C_1 can be transformed to C_2 using only flip and transpose operations in four steps as shown in Fig. 2: (a) we first transform C_1 to $\mathbb{C}_1$, (b) we transform C_2 to $\mathbb{C}_2$, (c) if $\mathbb{C}_1 \neq \mathbb{C}_2$, we then transform $\mathbb{C}_1$ to $\mathbb{C}_2$, and (d) finally, we reverse the steps of (b) to transform $\mathbb{C}_2$ to C_2. Effectively, we transform C_1 to $\mathbb{C}_1$ to $\mathbb{C}_2$ to C_2 as shown in Fig. 2.

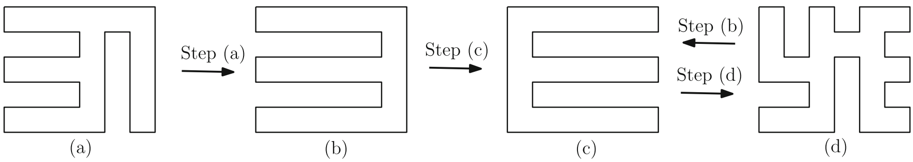

Fig. 2. (a) C_1, (b) $\mathbb{C}_1$, (c) $\mathbb{C}_2$, and (d) C_2. Steps (a) and (b): Transforming 1-complex Hamiltonian cycles C_1 and C_2 to canonical Hamiltonian cycles $\mathbb{C}_1$ and $\mathbb{C}_2$, respectively. Step (c): Transforming $\mathbb{C}_1$ to $\mathbb{C}_2$. Step (d): Reversing Step (b).

We first assume that C_1 has only three types of cookies and give an algorithm to transform C_1 to a canonical Hamiltonian cycle in Sect. 3.1. In Sect. 3.2, we use the algorithm in Sect. 3.1 to design an algorithm for the general case, where C_1 can have four types of cookies (Steps (a), (b) and (d)). In Sect. 3.3, we give an algorithm for transforming any canonical Hamiltonian cycle to another canonical Hamiltonian cycle (Step (c)). Finally, in Sect. 3.4, we analyze the time complexity of the algorithms.

3.1 1-Complex Hamiltonian Cycles with Three Types of Cookies

In this section, we give an algorithm, called *Algorithm* 1, to transform a 1-complex Hamiltonian cycle with at most three types of cookies to a canonical Hamiltonian cycle. We first give an overview of Algorithm 1.

Let C be a 1-complex Hamiltonian cycle on an $m \times n$ grid graph G such that C has at most three types of cookies. Since we can rotate G by multiples of 90°, we can assume that C has no east cookies. Thus all the edges on the east boundary, which is also column $n-1$, must be on C. The only bends are at the corner vertices $v_{n-1,0}$ and $v_{n-1,m-1}$. We therefore start from column $n-2$ and scan the vertices from top (y-coordinate 0) to bottom (y-coordinate $m-1$) in that column. Scanning down column $n-2$ from the first internal vertex $v_{n-2,1}$ toward the last one $v_{n-2,m-2}$ we note that these internal vertices must be covered by a north cookie until the first encounter with a west or south cookie. After a west cookie is met, subsequent internal vertices must be covered by west cookies until either a south cookie is met or the south boundary is reached. If a south cookie is met, it contains all remaining internal vertices. Note that any north and south cookies covering the internal vertices of column $n-2$ must be on the vertical track t_{n-3}. Let the height of the north and south cookies in vertical track t_{n-3} be y_N and y_S, respectively, where $y_N = 0$ if there is no north cookie in this track and $y_S = 0$ if there is no south cookie in this track. Then, the west cookies we encounter in column $n-2$ must be in horizontal tracks $t_{y_N+1}, t_{y_N+3}, \ldots, t_{m-3-y_S}$.

We apply transpose operations on the west cookies until there are no west cookies in the vertical track t_{n-3} and the height of the north cookie is $m-2-y_S$. Then we apply flip on the south cookie until the height of the north cookie is maximum (i.e., $m-2$). An example is shown in Fig. 3. We then move two columns to the left and continue the same process until we reach column 1 or column 0 (the west boundary). Observe that at any time, the internal vertices in the vertical tracks that have already been processed are covered by north cookies.

If n is even, we reach column 0 eventually. Then the algorithm ends as the Hamiltonian cycle now has only north cookies, which means that we have obtained a canonical Hamiltonian cycle. Otherwise, we reach column 1. In this case the internal vertices in column 1 must be covered by only west cookies, since a north or south cookie requires two columns. We then apply $(n-3)/2$ transpose operations on each west cookie, starting from the bottommost (in the horizontal track t_{m-3}) west cookie, and get a canonical Hamiltonian cycle with only west cookies. We now prove the correctness of Algorithm 1.

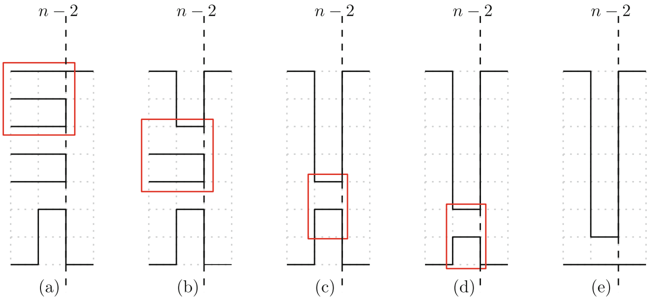

Fig. 3. (a) Column $n-2$ intersects two west cookies and a south cookie. Apply transpose in the 3×3 subgrid inside the red box, where the top left corner of the subgrid is the pin vertex, (b) apply another transpose, (c) then a flip in the 3×2 subgrid inside the red box, (d) apply another flip, and reach (e) final state. (Color figure online)

Theorem 1. *Let C be a 1-complex Hamiltonian cycle on an $m \times n$ grid graph G with no east cookies. Then Algorithm 1 transforms C to a canonical Hamiltonian cycle $\mathbb{C}$ of G, where $\mathbb{C}$ has either north cookies or west cookies.*

Proof. Since C is a Hamiltonian cycle, at least one of m and n must be even [8]. If n is even, we eventually reach column 0, the west boundary, and all the internal vertices in the columns between the east and west boundaries are covered by north cookies only. We now assume that n is odd, and hence m must be even. When we reach column 1, all the internal vertices in columns 2 through $n-2$ are covered by only north cookies, so the internal vertices on column 1 must be covered by only west cookies. We then apply $(n-3)/2$ transposes on each west cookie. Since m is even and we are moving two rows up after each step, we will reach row 0, at which stage we will have a canonical Hamiltonian cycle with only west cookies. □

3.2 1-Complex Hamiltonian Cycles with Four Types of Cookies

In this section, we give an algorithm to transform any 1-complex Hamiltonian cycle to a canonical Hamiltonian cycle.

Let C be a 1-complex Hamiltonian cycle on a grid graph G with as many as four types of cookies. A *subpath* P' of C is a sequence of consecutive vertices on C such that P' is a Hamiltonian path in an $m' \times n'$ subgrid G' of G, where the two endpoints of P' are corner vertices of G' on the same row or same column. Without loss of generality, assume that the endpoints of P' are on row 0 of G'. We create a *subproblem* C' for C by adding another row above row 0 of G' and connecting the row's endpoints to the endpoints of P' to form a Hamiltonian cycle in that augmentation of G'. An example is shown in Fig. 4.

We first observe some properties of C.

Properties. Assume there exists a column x that intersects an east and a west cookie. Then there cannot exist any row that intersects both a north cookie

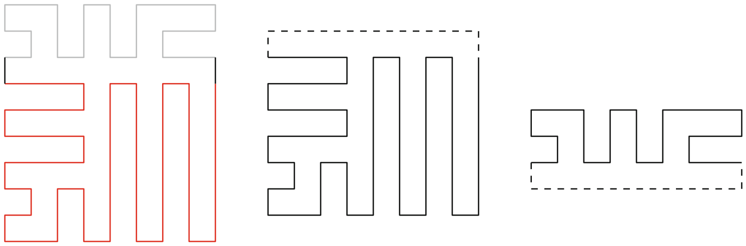

Fig. 4. (a) A 1-complex Hamiltonian cycle C. Two subpaths of C are shown in red and gray. (b) A subproblem for C created from the red subpath, and (c) a subproblem for C created from the gray subpath. The dashed lines show the edges that were added to the subpaths to create the subproblems. (Color figure online)

and a south cookie. Moreover, there must exist a pair of east and west cookies intersecting column x such that if the east cookie is on horizontal track t_y, then the west cookie must be on horizontal track t_{y+2} or t_{y-2}.

Let $C_1, C_2, \ldots, C_p$ be suproblems for C and let $P_1, P_2, \ldots, P_p$ be the corresponding subpaths. We say that C is *partitioned* into subproblems $C_1, C_2, \ldots, C_p$ if each vertex of C appears on exactly one P_i, $1 \leq i \leq p$. We now use the above properties to show that C can be partitioned into at most two subproblems, each with at most three types of cookies.

Lemma 1. *Let C be a* 1*-complex Hamiltonian cycle. Then C can be partitioned into at most two disjoint subproblems such that each subproblem has at most three types of cookies.*

Proof. We start scanning from the east boundary. If all the edges on that boundary are on C, then there are no east cookies. Therefore, C is a subproblem in itself with at most three types of cookies. Otherwise, there is at least one east cookie. We move one column to the left and continue scanning until one of the two following events occurs.

1. **A column intersects both east and west cookies.** Then by the properties of Hamiltonian cycles, there is no row that intersects both a north and a south cookie, and there exists a pair of east west cookies such that if the east cookie is in horizontal track t_y then the west cookie must be in horizontal track t_{y+2} or horizontal track t_{y-2}. Without loss of generality, we assume that the west cookie is in horizontal track t_{y-2} as shown in Fig. 5(a). Then there is no north cookie that goes below row $y-1$ and no south cookie that goes above row y. Let P' be the subpath of C from vertex $v_{0,y}$ to $v_{n-1,y}$ that includes the vertex $v_{0,m-1}$, and let P'' be the subpath of C from vertex $v_{0,y-1}$ to $v_{n-1,y-1}$ that includes the vertex $v_{0,0}$ as shown in Fig. 5(b). We then create two subproblems C' and C'' as shown in Fig. 5(c).

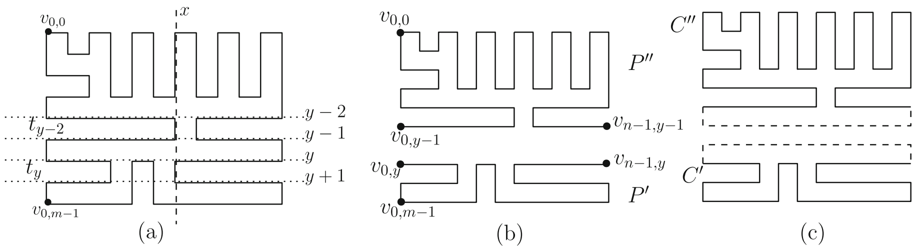

Fig. 5. (a) Column x intersecting both east and west cookies. (b) Subpaths P' and P'', (c) the two subproblems C' and C'' created from P' and P'', respectively.

2. **A column has no east cookies.** Let that column be x. There can be at most one south cookie and at most one north cookie intersecting column x. We now have the following cases to consider.

(a) Any north and south cookies that x intersects are in vertical track t_{x-1} as shown in Fig. 6(a). Let P' be the subpath of C from vertex $v_{x,0}$ to $v_{x,m-1}$ that includes the vertex $v_{0,0}$, and let P'' be the subpath of C from vertex $v_{x+1,0}$ to $v_{x+1,m-1}$ that includes the vertex $v_{n-1,0}$ (Fig. 6(b)). Then the subproblem created from P' will have no east cookies and the subproblem created from P'' will have no west cookies.

(b) Any north and south cookies are in vertical track t_x as shown in Fig. 6(c). Since this is the first column that intersects no east cookies when scanning to the left from the east boundary, there is at least one east cookie intersecting column $x+1$. Let the first east cookie from the top be in horizontal track t_y, where $0 < y \leq m-3$. Then there must be a west cookie in the same track, since the vertex $v_{x,y}$ cannot be covered by any north or south cookies. Then no north cookie goes below row $y-1$ and no south cookie comes above row $y+2$. Let P' be the subpath of C from vertex $v_{0,y-1}$ to $v_{n-1,y-1}$ that

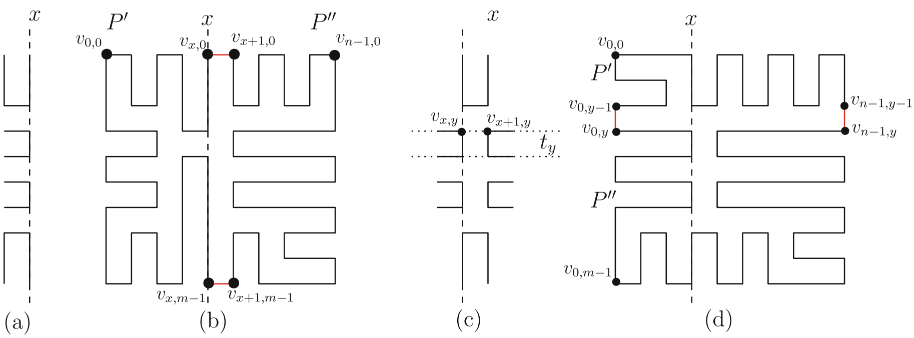

Fig. 6. (a) Vertical track t_{x-1} contains north or south cookies intersected by column x, (b) example of this case showing subpaths P' and P''. (c) Vertical track t_x contains north or south cookies intersected by column x, (d) example of this case.

includes the vertex $v_{0,0}$, and let P'' be the subpath of C from vertex $v_{0,y}$ to $v_{n-1,y}$ that includes the vertex $v_{0,m-1}$ (Fig. 6(d)). Then the subproblem created from P' will have no south cookies and the subproblem created from P'' will have no north cookies.

(c) Column x intersects north and south cookies on different vertical tracks. Let the height of the cookie in vertical track t_x be y. We assume that the north cookie is in vertical track t_x as shown in Fig. 7(a), where possibly column x intersects some west cookies. The case when the south cookie is in vertical track t_x is similar. Column $x+1$ must intersect an east cookie but no west cookies, so vertex $v_{x+1,y+1}$ must be covered by an east cookie. From x we move to the left column by column until we find a west cookie which must occur at some column $x' > 0$. There may be three cases as described below; see Figs. 7(b)–(d). If there is a west cookie in column x (Fig. 7(b)), then there must be one in horizontal track t_{y+1}, as the height of the north cookie is y. If there is no west cookie in column x, then since x does not intersect any east cookie, vertices $v_{x,y+1}$ to $v_{x,m-1}$ must lie in a south cookie or on the south boundary. Thus no column left of x intersects an east cookie, and thus the vertex $v_{x-1,y}$ is covered by a north cookie or a west cookie. Vertex $v_{x-1,y+1}$ must be covered by a south cookie.

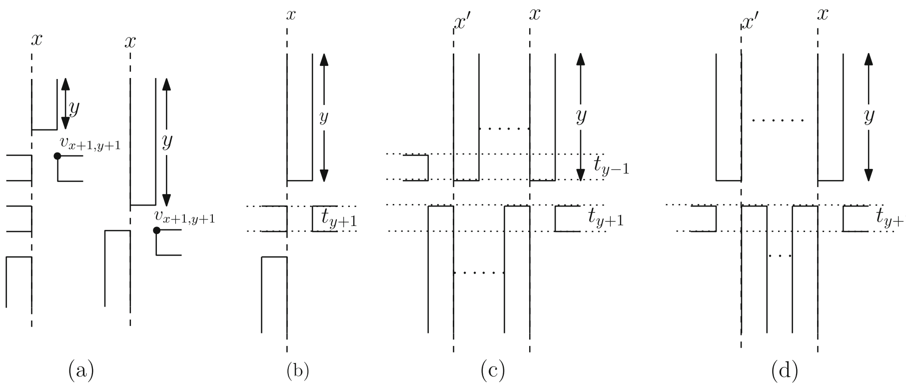

Fig. 7. (a) The north cookie is in vertical track t_x and the south cookie is in vertical track t_{x-1}. (b) Column x intersects a west cookie in horizontal track t_{y+1}. (c) The first west cookie to the left of column x is in horizontal track t_{y-1}. (d) The first west cookie to the left of column x is in horizontal track t_{y+1}.

We keep moving to the left until we arrive at a column x' such that there is a west cookie in column $x'-1$. Observe that all the north cookies between (and including) columns x' and x have height y and all the south cookies between (and including) these columns have height $m-2-y$. If there is a north cookie in vertical track $t_{x'}$, then the west cookie will be in horizontal track t_{y-1} as shown in Fig. 7(c). Otherwise, there is a north cookie in vertical track $t_{x'-1}$, and the west cookie will be in horizontal track t_{y+1} as shown in Fig. 7(d). Since we have a west cookie in either horizontal track t_{y-1} or t_{y+1} with height $x'-1$ and an east cookie in horizontal track t_{y+1} with height $n-2-x$, and all the north cookies in columns x' to x have height y, there is no north cookie in C with height greater than y. Similarly, there is no south cookie in C that has height greater than $m-2-y$.

We then create two subproblems in a similar way to that shown in Figs. 6(c)–(d). Let P' be the subpath of C from vertex $v_{0,y}$ to $v_{n-1,y}$ that includes the vertex $v_{0,0}$, and let P'' be the subpath of C from vertex $v_{0,y+1}$ to $v_{n-1,y+1}$ that include the vertex $v_{0,m-1}$. Then the subproblem C' created from P' will have no south cookies and the subproblem C'' created from P'' will have no north cookies.

Therefore, C can always be partitioned into at most two subproblems, each with at most three types of cookies. Observe that in each case above, the subpaths are connected on C by two edges on the boundary. □

We now give an algorithm to transform any 1-complex Hamiltonian cycle C to a canonical Hamiltonian cycle. We call it Algorithm 2.

We first partition C into at most two subproblems C' and C'' using Lemma 1. Let C' and C'' correspond to the subpaths P' and P'' of C, respectively. By the proof of Lemma 1, P' and P'' must be connected by two boundary edges e_1 and e_2. Then we use Algorithm 1 to transform C' and C'' to canonical Hamiltonian cycles. We then remove the extra rows or columns we added to P' and P'' to create C' and C'', respectively, and join the two subpaths using e_1 and e_2. Let C_1 be the Hamiltonian cycle obtained in this way. Then C_1 has at most two types of cookies. We apply Algorithm 1 on C_1 to obtain a canonical Hamiltonian cycle. We now prove correctness of Algorithm 2.

Theorem 2. *Let C be a 1-complex Hamiltonian cycle on an $m \times n$ grid graph G. Then Algorithm 2 transforms C to a canonical Hamiltonian cycle of G.*

Proof. By Lemma 1, C can be partitioned into at most two subproblems C' and C'', corresponding to subpaths P' and P'' of C, respectively. We then apply Algorithm 1 on C' and C'' to obtain canonical Hamiltonian cycles. Recall that Algorithm 1 takes a 1-complex Hamiltonian cycle with no east cookies and transforms it to a Hamiltonian cycle with only north cookies or only west cookies. Thus the canonical Hamiltonian cycles obtained from C' and C'' cannot have cookies from the boundaries that were added to P' and P''; removing those boundaries gives two subpaths that can be joined to obtain another 1-complex Hamiltonian cycle C_1 on G. Since C was partitioned into at most two subproblems, C_1 can have at most two types of cookies. Now we can apply Algorithm 1 on C_1 again to obtain a canonical Hamiltonian cycle of G. □

3.3 Transformation Between Canonical Hamiltonian Cycles

In this section we give an algorithm to transform a canonical Hamiltonian cycle $\mathbb{C}_1$ of an $m \times n$ grid graph G to another canonical Hamiltonian cycle $\mathbb{C}_2$ of G. We call it Algorithm 3.

Without loss of generality, assume that $\mathbb{C}_1$ has west cookies. If $\mathbb{C}_2$ has east cookies, we apply $n-2$ flips on the west cookie in horizontal track t_y, $1 \leq y \leq m-3$, starting from horizontal track t_1 until we have only an east cookie in that track. If $\mathbb{C}_2$ has north (or south) cookies, we apply $(m-2)/2$ transposes in the vertical tracks t_x, $1 \leq x \leq n-3$, such that the pin vertex is in column $x-1$, starting from vertical track t_{n-3} until that track is covered by only one north (south) cookie.

3.4 Complexity Analysis

In this section we analyze the time complexity of Algorithms 1, 2 and 3.

Let C be a 1-complex Hamiltonian cycle in an $m \times n$ grid graph G. In Algorithm 1, we scan every column of G from column $n-2$ to column 1 or column 0. The scanning takes linear time $O(mn)$. We apply at most $m-2$ transposes and flips in each vertical track of C. Since there are $(n-2)/2$ vertical tracks, at most $(m-2)(n-2)/4$ operations are performed on the vertical tracks. In case n is odd, $(n-3)/2$ transposes are applied on each horizontal track. Since there are $(m-2)/2$ horizontal tracks, exactly $(m-2)(n-3)/4$ additional transpose operations are performed in this case. Therefore, the algorithm takes $O(mn)$ time in total, since each flip or transpose operation takes $O(1)$ time.

In Algorithm 2, partitioning C into two subproblems and merging the solutions to the subproblems take $O(mn)$ time. We apply Algorithm 1 on the two subproblems and finally on the cycle obtained from merging the solutions to the subproblems. Therefore, the total running time of Algorithm 2 is $O(mn)$.

In Algorithm 3, we apply $O(mn)$ flips or transposes, which takes $O(mn)$ time. Since Algorithm 2 (Steps (a) and (c)) takes $O(mn)$ time and Algorithm 3 (Step (b)) takes $O(mn)$ time, we can transform any 1-complex Hamiltonian cycle on G to any other 1-complex Hamiltonian cycle on G in $O(mn)$ time with $O(mn)$ flips and transpose operations, in other words, in linear time (and space).

4 Conclusions

In this paper, we have studied Hamiltonian cycles on grid graphs, by which we mean solid rectangular grid graphs. We defined a complexity measure for Hamiltonian cycles on rectangular grid graphs, and two local operations, flip and transpose. We showed that any 1-complex Hamiltonian cycle can be transformed to any other 1-complex Hamiltonian cycle using only a linear number of flips and transposes, thereby initiating a study of k-complex Hamiltonian cycles in grid graphs as we next describe.

We define *Hamiltonian cycle graph* $\mathcal{G}_{m,n}$ to be the graph whose vertices are Hamiltonian cycles on an $m \times n$ grid graph G. Two vertices u, v of $\mathcal{G}_{m,n}$ have an edge between them if one can obtain u from v and vice versa by applying a single flip or transpose operation. The subgraph $\mathcal{G}_{m,n,k}$ of $\mathcal{G}_{m,n}$ contains exactly the Hamiltonian cycles with bend complexity k. Our result shows that $\mathcal{G}_{m,n,1}$ is a connected graph and that the diameter of $\mathcal{G}_{m,n,1}$ is at most $O(mn)$. We pose the question whether $\mathcal{G}_{m,n,k}$ is a connected graph, where $k > 1$. We also ask whether similar results can be achieved for grid graphs with non-rectangular outer face boundary or grid graphs with holes. It would also be interesting to find out whether our results can be extended to Hamiltonian paths.

References

1. Afrati, F.: The hamilton circuit problem on grids. RAIRO Theor. Inform. Appl. **28**(6), 567–582 (1994)
2. Arkin, E.M., Fekete, S.P., Islam, K., Meijer, H., Mitchell, J.S., Rodríguez, Y.N., Polishchuk, V., Rappaport, D., Xiao, H.: Not being (super)thin or solid is hard: a study of grid hamiltonicity. Comput. Geom. **42**(6–7), 582–605 (2009)
3. Bose, P.: Flips. In: Didimo, W., Patrignani, M. (eds.) GD 2012. LNCS, vol. 7704, pp. 1–1. Springer, Heidelberg (2013). doi:10.1007/978-3-642-36763-2_1
4. Cho, H.-G., Zelikovsky, A.: Spanning closed trail and hamiltonian cycle in grid graphs. In: Staples, J., Eades, P., Katoh, N., Moffat, A. (eds.) ISAAC 1995. LNCS, vol. 1004, pp. 342–351. Springer, Heidelberg (1995). doi:10.1007/BFb0015440
5. des Cloizeaux, J., Jannik, G.: Polymers in Solution: Their Modelling and Structure. Clarendon Press, Oxford (1987)
6. Gorbenko, A., Popov, V.: On hamilton paths in grid graphs. Adv. Stud. Theor. Phys. **7**(3), 127–130 (2013)
7. Gorbenko, A., Popov, V., Sheka, A.: Localization on discrete grid graphs. In: He, X., Hua, E., Lin, Y., Liu, X. (eds.) CICA 2011. Lecture Notes in Electrical Engineering, vol. 107, pp. 971–978. Springer, Dordrecht (2012). doi:10.1007/978-94-007-1839-5_105
8. Itai, A., Papadimitriou, C.H., Szwarcfiter, J.L.: Hamilton paths in grid graphs. SIAM J. Comput. **11**(4), 676–686 (1982)
9. Umans, C., Lenhart, W.: Hamiltonian cycles in solid grid graphs. In: Proceedings of the 38th Annual Symposium on Foundations of Computer Science, FOCS 1997, pp. 496–505. IEEE Computer Society (1997)
10. Wagner, K.: Bemerkungen zum vierfarbenproblem. Jahresbericht der Deutschen Mathematiker-Vereinigung **46**, 26–32 (1936)